RISHITHA CHILAKA

AI/ML Engineer | RAG & LLM Applications | Applied Machine Learning

Houston, TX | (628) 724-1023 | rishithareddych21@gmail.com |

PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience in machine learning and data analytics, including hands-on work building production Generative AI and RAG systems. Contributed to retrieval pipelines, LLM evaluation frameworks, and hallucination mitigation as part of an enterprise clinical-intelligence platform, with additional background in document classification, NLP, and data-driven analytics. Hands-on with Python, SQL, PyTorch, Azure OpenAI, Hugging Face Transformers, pgvector, vector embeddings, re-ranking, MLflow, GitHub Actions, Docker, and model monitoring

TECHNICAL SKILLS

Languages & Data: Python, SQL, Bash, Git, Pandas, NumPy **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Large Language Models(LLMs), Re-ranking, Vector embeddings, Semantic search, Prompt engineering, Azure OpenAI, Hugging Face Transformers, Hallucination mitigation **LLM Evaluation:** Golden datasets, Evaluation harnesses, Retrieval recall@k, Groundedness/Faithfulness evaluation, Regression testing, Error analysis, Prompt versioning **Machine Learning:** PyTorch, Scikit-learn, NLP, Document Classification, Feature Engineering, Classification, model evaluation **AI/ML systems & MLOPs:** Vector Databases, pgvector, Document Processing, OCR, Retrieval Pipelines, MLflow, CI/CD, GitHub Actions, Docker, Model Monitoring, Model Versioning, Rollback **Cloud:** Microsoft Azure

PROFESSIONAL EXPERIENCE

AI/ML Engineer — Optum — Houston, TX (Remote) | Jan 2025 - Present

- Designed and built production components of a Retrieval-Augmented Generation (RAG) enterprise intelligence platform using Azure OpenAI, Hugging Face Transformers, and vector embedding pipelines to support retrieval and synthesis across 40,000+ enterprise documents and structured/unstructured data.
- Built a distributed Python-based ETL and document-processing pipeline for parsing, OCR fallback, section-aware chunking, embeddings, and incremental re-indexing in pgvector, supporting ingestion and updates across a large enterprise document corpus
- Improved retrieval Recall@10 from 0.58 to 0.79 by restructuring chunks around document section headings and adding a re-ranking stage, addressing retrieval quality as a key bottleneck to answer quality.
- Integrated a golden-set evaluation framework into GitHub Actions CI/CD to measure groundedness and Recall@K for prompt and retrieval changes, catching 3 regressions before production and establishing a repeatable quality gate.
- Reduced hallucination rate by 50% on the evaluation set by routing out-of-corpus questions to analysts instead of generating unsupported responses, improving production answer reliability.
- Reduced inference cost for simpler queries by routing single-document lookups to a smaller model and multi-document comparisons to a larger model, with evaluation showing no measurable quality degradation on the simpler workload.
- Improved document-classification F1 from 0.61 to 0.83 across 14 document classes by developing and productionizing a PyTorch based classifier, replacing keyword-rule routing for enrollment paperwork.
- Implemented prompt versioning and rollback controls after tracing a two-day answer-quality regression to an untracked prompt change, adding change control and recovery capabilities to the production AI workflow.
- Monitored production model performance and prediction quality, supporting detection of performance degradation and informing model updates.

Machine Learning Engineer — IBM — Bangalore | Oct 2020 - Mar 2023

- Assisted in developing anomaly detection models for log data processing 450K+ logs per hour, contributing to a 32% reduction in false positives.
- Developed and evaluated NLP classification models using spaCy, improving classification accuracy by 26%.
- Supported optimization of TensorFlow and CUDA-based inference pipelines to improve model response time.
- Implemented data drift monitoring processes to help maintain model stability and performance.
- Assisted with deploying and maintaining containerized ML services using Python and machine learning technologies.

EDUCATION

Union Commonwealth University — Master of Science, Management Information Systems — Apr 2023 - Dec 2024